# FINM3123 Introduction to Econometrics

## Project

**Due date of report: 11:59pm on December 14, 2024**
**Submit through iSpace (turnitin).**

Each group will conduct the specific econometric analysis and write on their own a term paper of about 5-10 pages. You should write the results in a way that is easy-to-read, and interpret your findings clearly.

**There are some potential topics you may consider:**

1. What factors affect the performance of professional athletes, such as NBA, CBA, etc.? For each player, we have information on points scored, rebounds, assists, playing time, marital status, and demographics. You can use multiple regression analysis to test whether the productivity measures differ with these factors. Annual earnings can be used to directly reflect productivity.
2. What factors affect student performance at universities? Does spending per student matter? Do student-teacher ratios have any effects? It is difficult to estimate ceteris paribus effects because spending is related to other factors, such as family incomes or poverty rates.
3. Are cigarette smokers less productive? A variant on this is: Do workers who smoke take more sick days (everything else being equal)? You could use data at, say, the metropolitan level. Something like average productivity in manufacturing can be related to percentage to manufacturing workers who smoke. Other variables, such as average worker education, capital per worker, and size of the city (you can think of more), should be controlled for.
4. What factors affect manufacturing productivity at the province/city level? In addition to levels of the capital and worker education, you could look at other factors.
5. Use provincial- or city-level data to look at the factors that affect education spending per pupil. An interesting question is: Other things being equal (such as income and education levels of residents), do districts with a larger percentage of elderly people spend less on schools?
6. Other topics that you are interested. Once you choose this option, you may want to discuss

with the course instructor to make sure your topic is doable.

Once you select the topic, you should collect your own data, and then write a more substantial term paper. In your paper, you must report the data sources clearly.

While writing styles vary, many papers follow the same general outline. The following paragraphs include ideas for section headings and explanations about what each section should contain. They are only suggestions and hardly need to be strictly followed. In the final paper, each section would be given a number, usually starting with one for the introduction.

**Introduction**

The introduction states the basic objectives of the study and explains why it is important. It generally entails a review of the literature, indicating what has been done and how previous work can be improved upon. An extensive literature review can be put in a separate section. Most researchers like to summarize the findings of their paper in the introduction. But the summary should not be too involved because neither the methods nor the data used to obtain the estimates have yet been introduced.

**Econometric Models and Estimation Methods**

It is very useful to have a section that contains a few equations of the sort you estimate and present in the results section of the paper. After specifying a model or models, it is appropriate to discuss estimation methods. Any assumptions that are used in obtaining an estimable econometric model should be clearly discussed.

**The Data**

You should always have a section that carefully describes the data used in the empirical analysis. In particular, all applicable public data sources should be included in the references (or footnotes), and short data sets can be listed in an appendix. If you used your own survey to collect the data, a copy of the questionnaire should be presented in an appendix. Along with a discussion of the data sources, be sure to discuss the units of each of the variables. Including a table of variable definitions is very useful to the reader. It is also very informative to present a table of summary statistics, such as minimum and maximum values, means, and standard deviations for each variable. You should always clearly state how many observations you have.

## Results

The results section should include your estimates of any models formulated in the models section. If only a few equations are estimated, you can present the results in equation form with standard errors in parentheses below estimated coefficients. If your model has several explanatory variables and you are presenting several variations on the general model, it is better to report te results in tabular rather than equation form. The most important thing is to discuss the interpretation and strength of your empirical results. In addition to discussing estimates from the most general model, you can provide interesting special cases, especially those needed to test multiple hypotheses.

## Conclusions

This can be a short section that summarizes what you have done. The conclusion should also discuss caveats to the conclusions drawn, and it might even suggest directions for further research.